INF1004 Linear Algebra
(Part II)
Lecture 4: Eigenvectors and Eigenvalues

Lecture Outline

- Eigenvalues & Eigenvectors
- Eigen decomposition / Diagonalization
- Elementary Functions of a Matrix
- Application of Eigen decomposition in Solving of Differential Equations

Recap: Linear Transformation

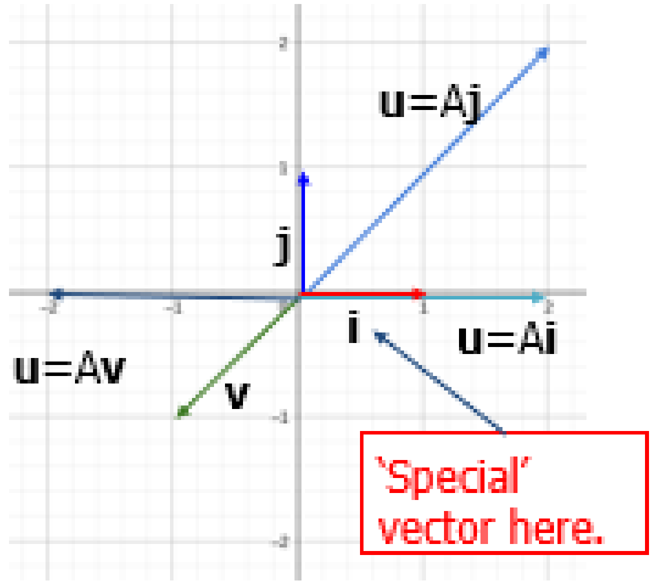
- A transformation from one vector space ***V*** to another vector space ***U***
- A rule ***T*** that assigns each vector ***v*** in ***V*** to a unique vector ***T(v)*** in ***U.***
- T: $\boldsymbol{v} \mapsto \boldsymbol{u}$

Recap: Linear Transformation

For example, the shear + scale transformation:

$$\mathbf{u} = A\mathbf{v} = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} \mathbf{v}$$

From the graph, it looks like most input vectors are rotated by matrix A. However, notice that the vector $[1\ 0]^T$ remains on its span.



Linear Transformation (Eigen Values and Eigen vectors)

Let $A = \begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$, $\mathbf{y} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

What happens with x and y if they are transformed by A?

$$A\mathbf{x} = \begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ -6 \end{pmatrix} \qquad A\mathbf{y} = \begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

What we observed here is

$$A\mathbf{x} = \begin{pmatrix} 4 \\ -6 \end{pmatrix} = -2\begin{pmatrix} -2 \\ 3 \end{pmatrix} = -2\mathbf{x}$$

$$A\mathbf{y} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1\mathbf{y}$$

Linear Transformation (Eigen Values and Eigen vectors)

$A\mathbf{x} = -2\mathbf{x}$ $\qquad\qquad$ $A\mathbf{y} = 1\mathbf{y}$

When we operate on the vector **x** with the matrix A, instead of getting a different vector (as we would normally do), we get the same vector **x** multiplied by some constant.

We call the values **-2** and **1** the eigenvalues of the matrix A,
And the vectors **x** and **y** are called eigenvectors for the matrix A.

More Formal Definition

An eigenvector of an ***n x n*** matrix ***A*** is a non zero vector such that ***Ax* = *λx*** for some scalar ***λ***. ***λ*** is called the eigenvalue of ***A*** if there is any nontrivial solution ***x*** of ***Ax* = *λx*** such that ***x*** is called the eigenvector corresponding to ***λ***

Visualizing Matrix-vector multiplication

Visualizing matrix-vector multiplication

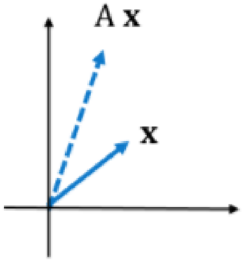
When you multiply a matrix and a vector, the result is another vector.

In this way, the matrix A can be thought of as a transformation that turns $\mathbf{x}$ into $\mathbf{y}$.

matrix

$$A\,\mathbf{x} = \mathbf{y}$$

vectors



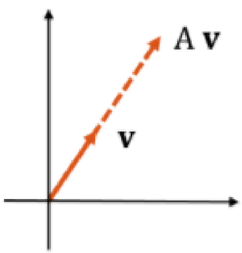
When you multiply a matrix and its eigenvector, it's as though the eigenvector was just multiplied by a number.

The eigenvector is the direction that isn't changed when multiplied by A - it's just scaled, and the scaling factor is the eigenvalue.

matrix eigenvalue

$$A\,\mathbf{v} = \lambda\,\mathbf{v}$$

eigenvector



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Linear Transformation (Eigen Values and Eigen vectors)

A scalar λ is called an **eigenvalue** of A if there is a nontrivial solution **x** of $A\mathbf{x} = \lambda\mathbf{x}$; such an **x** is called an **eigenvector** corresponding to λ .

Non-trivial: having some variables or terms that are not equal to zero or an identity.

Example

Let $A = \begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}$ and $\mathbf{u} = \begin{pmatrix} 6 \\ -5 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$

Are **u** and **v** eigenvectors of A?

Example

$$A = \begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix} \qquad \mathbf{u} = \begin{pmatrix} 6 \\ -5 \end{pmatrix} \qquad \mathbf{v} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

$$A\mathbf{u} = \begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}\begin{pmatrix} 6 \\ -5 \end{pmatrix} = \begin{pmatrix} -24 \\ 20 \end{pmatrix} = -4\begin{pmatrix} 6 \\ -5 \end{pmatrix} = -4\mathbf{u}$$

Thus, **u** is an eigenvector corresponding to an eigenvalue -4.

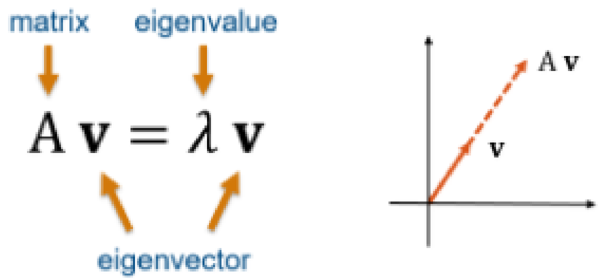
$$A\mathbf{v} = \begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}\begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} -9 \\ 11 \end{pmatrix} \neq \lambda\begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

Thus, **v** is not an eigenvector of A, because A**v** is not a multiple of **v**.

MATLAB on Eigenvectors and Eigenvalues

Visualizing Eigenvectors

When you multiply a matrix and its eigenvector, the eigenvector is simply scaled, and the scaling factor is the eigenvalue.



The `eig` Function

```
>> [V,D] = eig(A)
```

Outputs

`V`	Matrix whose columns are the eigenvectors of `A`. The eigenvectors are normalized to have magnitude 1.
`D`	Diagonal matrix whose diagonal elements are the eigenvalues of `A`.

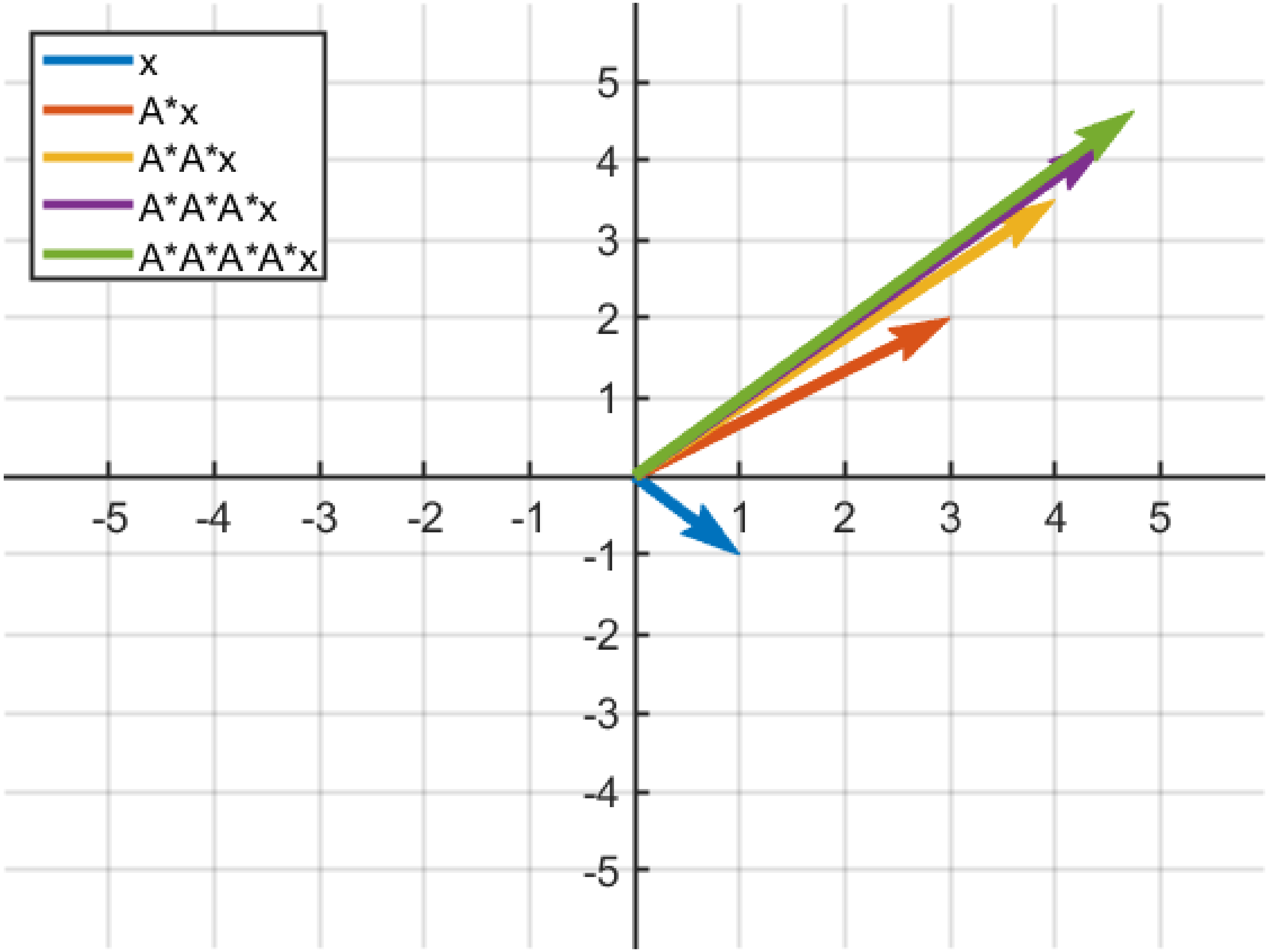
Inputs

`A`	The matrix for which to find the eigenvalue decomposition.

For each eigenvector in `V`, its corresponding eigenvalue can be found in the same column of the matrix `D`.

Example

Multiplying $A*A*A*...*A*\mathbf{x}$, given the plot below, which of the following vectors is an eigenvector of A?



Characteristic Equation

$A\mathbf{x} = \lambda\mathbf{x} \quad = \lambda I\mathbf{x}$, where **I** is an identity matrix

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

By definition, **x** must be non-zero vector.

Hence, for that to have a solution, determinant $|A - \lambda I| = 0$

$|A - \lambda I| = 0$ is called the **characteristic equation** of A.

The determinant $|A - \lambda I|$ is called the **characteristic determinant**.

The values of λ that set this determinant to 0 (the roots of the characteristic equation) are the eigenvalues of $\mathbf{A}$.

Example

Find the eigenvalues and eigenvectors of matrix $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Example

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix}$$

$$|A - \lambda| = \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - 1$$

The characteristic equation is $\lambda^2 - 1 = 0$

The roots are the eigenvalues of A: $\lambda = 1, \ \lambda = -1$

Example

The eigenvectors of A is any non-zero vector x that satisfies

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

Since there are 2 eigenvalues, there will be 2 eigenvectors.

For $\lambda = 1$ there is an eigenvector $\mathbf{x}^{(1)}$ such that $\boldsymbol{A}\boldsymbol{x}^{(1)} = \boldsymbol{x}^{(1)}$

$$(A - \lambda I)\mathbf{x}^{(1)} = \mathbf{0}$$

$$\begin{pmatrix} 0-1 & 1 \\ 1 & 0-1 \end{pmatrix} \begin{pmatrix} x_1^{(1)} \\ x_2^{(1)} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x_1^{(1)} \\ x_2^{(1)} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$-x_1^{(1)} + x_2^{(1)} = 0 \text{ and } x_1^{(1)} - x_2^{(1)} = 0$$

$$\text{Set } x_2^{(1)} = 1, \text{ then } x_1^{(1)} = 1.$$

Thus, the eigenvector corresponding to $\lambda = 1$ is $x^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Example

Similarly, for $\lambda = -1$ there is an eigenvector $\mathbf{x}^{(2)}$ such that $\mathbf{A}x^{(2)} = -x^{(2)}$

$$(A - \lambda I)\mathbf{x}^{(2)} = \mathbf{0}$$

$$\begin{pmatrix} 0-(-1) & 1 \\ 1 & 0-(-1) \end{pmatrix} \begin{pmatrix} x_1^{(2)} \\ x_2^{(2)} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Longrightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1^{(2)} \\ x_2^{(2)} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$x_1^{(2)} + x_2^{(2)} = 0 \text{ and } x_1^{(2)} + x_2^{(2)} = 0$$

Thus the eigenvector corresponding to $\lambda = -1$ is: $x^{(2)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

Class Exercise 1

Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$

Class Exercise 1 - Solution

$$A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} -2-\lambda & 1 \\ 1 & -2-\lambda \end{pmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} -2-\lambda & 1 \\ 1 & -2-\lambda \end{vmatrix} = (-2-\lambda)(-2-\lambda) - (1)(1)$$

$$= 4 + 4\lambda + \lambda^2 - 1 \quad = \lambda^2 + 4\lambda + 3$$

$$\lambda^2 + 4\lambda + 3 \quad = (\lambda+3)(\lambda+1) = 0$$

The eigenvalues are:

$$\lambda_1 = -3 \qquad \lambda_2 = -1$$

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Class Exercise 1 - Solution

To find the eigenvectors, we use

$$\lambda(\mathbf{A} + \mathbf{B}) = \lambda\mathbf{A} + \lambda\mathbf{B}$$

For λ = -3,
$$\begin{pmatrix} -2-(-3) & 1 \\ 1 & -2-(-3) \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$x_1 + x_2 = 0$$

$$x_1 = -x_2$$

Set $x_2 = 1$, then $x_1 = -1$

An eigenvector corresponding to eigenvalue -3 is $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

Class Exercise 1 - Solution

For λ = -1,

$$\begin{pmatrix} -2-(-1) & 1 \\ 1 & -2-(-1) \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$-x_1 + x_2 = 0$$

$$x_1 = x_2$$

Set $x_2 = 1$, then $x_1 = 1$

The eigenvector corresponding to eigenvalue -1 is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Class Exercise 2

Find the eigenvalues and eigenvectors of matrix

$$A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

Class Exercise 2 - Solution

$$A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1-\lambda & 1 & -2 \\ -1 & 2-\lambda & 1 \\ 0 & 1 & -1-\lambda \end{pmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 & -2 \\ -1 & 2-\lambda & 1 \\ 0 & 1 & -1-\lambda \end{vmatrix} = -\lambda^3 + 2\lambda^2 + \lambda - 2$$

The characteristic equation is $-\lambda^3 + 2\lambda^2 + \lambda - 2 = 0$

$$(1+\lambda)(1-\lambda)(\lambda-2) = 0$$

Class Exercise 2 - Solution

$$(1+\lambda)(1-\lambda)(\lambda-2)=0$$

The eigenvalues are: $\lambda_1 = 2,\ \lambda_2 = 1, \lambda_3 = -1$

The eigenvectors are obtained from

$$(A-\lambda I)e_i = \mathbf{0}$$

$$A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1-\lambda & 1 & -2 \\ -1 & 2-\lambda & 1 \\ 0 & 1 & -1-\lambda \end{pmatrix}\begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix} = 0$$

$$\rightarrow \quad \begin{array}{cccc} (1-\lambda)e_1 & +e_2 & -2e_3 & = 0 \\ -e_1 & +(2-\lambda)e_2 & +e_3 & = 0 \\ & +e_2 & -(1+\lambda)e_3 & = 0 \end{array}$$

Class Exercise 2 - Solution

When $\lambda = \lambda_1 = 2$,

$$\begin{aligned} (1-2)e_1 \quad &+e_2 \quad &-2e_3 &= 0 \\ -e_1 \quad &+(2-2)e_2 \quad &+e_3 &= 0 \\ &+e_2 \quad &-(1+2)e_3 &= 0 \end{aligned}$$

$$\rightarrow \quad \begin{aligned} -e_1 \quad +e_2 \quad -2e_3 &= 0 \quad (1) \\ -e_1 \qquad\quad +e_3 &= 0 \quad (2) \\ +e_2 \quad -3e_3 &= 0 \quad (3) \end{aligned}$$

$$(A - \lambda I)e_i = \begin{pmatrix} 1-2 & 1 & -2 \\ -1 & 2-2 & 1 \\ 0 & 1 & -1-2 \end{pmatrix} \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix}$$

From Eq. (2), we get $e_1 = e_3$

From Eq. (3), we get $e_2 = 3e_3$

Set $e_3 = 1$, then $e_2 = 3$, $e_1 = 1$

The eigenvector corresponding to λ= 2 is $\mathbf{e}^{(1)} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$.

Class Exercise 2 - Solution

When $\lambda = \lambda_2 = 1,$

$$\begin{aligned} (1-1)e_1 \quad &+e_2 \quad &-2e_3 &= 0 \\ -e_1 \quad &+(2-1)e_2 \quad &+e_3 &= 0 \\ &+e_2 \quad &-(1+1)e_3 &= 0 \end{aligned}$$

$$\rightarrow \quad \begin{aligned} &+e_2 \quad -2e_3 = 0 \\ -e_1 \quad &+e_2 \quad +e_3 = 0 \\ &+e_2 \quad -2e_3 = 0 \end{aligned}$$

(1)

(2)

(3)

Eq. (1) – Eq. (2), we get $e_1 = 3e_3$

From Eq. (3) or Eq. (1), we get $e_2 = 2e_3$

Set $e_3 = 1$, then $e_2 = 2$, $e_1 = 3$

The eigenvector corresponding to ?= 1 is $\mathbf{e}^{(2)} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$.

Class Exercise 2 - Solution

When $\lambda = \lambda_2 = -1,$

$$\begin{array}{rrrl} (1+1)e_1 & +e_2 & -2e_3 & = 0 \\ -e_1 & +(2+1)e_2 & +e_3 & = 0 \\ & +e_2 & -(1-1)e_3 & = 0 \end{array}$$

$$\rightarrow \quad \begin{array}{rrrl} 2e_1 & +e_2 & -2e_3 & = 0 \\ -e_1 & +3e_2 & +e_3 & = 0 \\ & +e_2 & & = 0 \end{array}$$

(1)

(2)

(3)

We get $e_2 = 0$ $\qquad e_1 = e_3$

Set e_3 = 1, then e_1 = 1

The eigenvector corresponding to λ= -1 is $\mathbf{e}^{(3)} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$.

Class Exercise 3

Find the eigenvalues and eigenvectors of matrix

$$A = \begin{pmatrix} 3 & -3 & 2 \\ -1 & 5 & -2 \\ -1 & 3 & 0 \end{pmatrix}$$

Class Exercise 3 - Solution

$$A = \begin{pmatrix} 3 & -3 & 2 \\ -1 & 5 & -2 \\ -1 & 3 & 0 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 3 & -3 & 2 \\ -1 & 5 & -2 \\ -1 & 3 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3-\lambda & -3 & 2 \\ -1 & 5-\lambda & -2 \\ -1 & 3 & -\lambda \end{pmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 3-\lambda & -3 & 2 \\ -1 & 5-\lambda & -2 \\ -1 & 3 & -\lambda \end{vmatrix} = -\lambda^3 + 8\lambda^2 - 20\lambda + 16$$

The characteristic equation is $-\lambda^3 + 8\lambda^2 - 20\lambda + 16 = 0$

$$-(\lambda - 4)(\lambda - 2)^2 = 0$$

Class Exercise 3 - Solution

$$-(\lambda-4)(\lambda-2)^2=0$$

The eigenvalues are: $\lambda_1=4,\ \lambda_2=\lambda_3=2$

The eigenvectors are obtained from

$$(A-\lambda I)e_i=\mathbf{0}$$

$$\begin{pmatrix}3-\lambda & -3 & 2\\ -1 & 5-\lambda & -2\\ -1 & 3 & -\lambda\end{pmatrix}\begin{pmatrix}e_1\\ e_2\\ e_3\end{pmatrix}$$
$$=0$$

$$\rightarrow \quad \begin{array}{ccc}(3-\lambda)e_1 & -3e_2 & +2e_3=0\\ -e_1 & +(5-\lambda)e_2 & -2e_3=0\\ -e_1 & +3e_2 & -\lambda e_3=0\end{array}$$

Class Exercise 3 - Solution

When $\lambda = \lambda_1 = 4$,

$$\begin{aligned} (3-4)e_1 \quad & -3e_2 \quad & +2e_3 = 0 \\ -e_1 \quad & +(5-4)e_2 \quad & -2e_3 = 0 \\ -e_1 \quad & +3e_2 \quad & -4e_3 = 0 \end{aligned}$$

$$\rightarrow \quad \begin{aligned} -e_1 \quad & -3e_2 \quad & +2e_3 = 0 \qquad (1) \\ -e_1 \quad & +e_2 \quad & -2e_3 = 0 \qquad (2) \\ -e_1 \quad & +3e_2 \quad & -4e_3 = 0 \qquad (3) \end{aligned}$$

Eq(1) – Eq(2), we get $-2e_1 - 2e_2 = 0$

Eq(1) + Eq(3), we get $-2e_1 - 2e_3 = 0$

Set $e_1 = 1$, then $e_2 = -1$, $e_3 = -1$

The eigenvector corresponding to eigenvalue 4 is $\mathbf{e}^{(1)} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$.

Class Exercise 3 - Solution

When $\lambda = \lambda_2 = \lambda_3 = 2$,

$$\begin{aligned} (3-2)e_1 \quad & -3e_2 \quad & +2e_3 = 0 \\ -e_1 \quad & +(5-2)e_2 \quad & -2e_3 = 0 \\ -e_1 \quad & +3e_2 \quad & -2e_3 = 0 \end{aligned}$$

$$\rightarrow \quad \begin{aligned} e_1 \quad & -3e_2 \quad & +2e_3 = 0 \quad & (1) \\ -e_1 \quad & +3e_2 \quad & -2e_3 = 0 \quad & (2) \\ -e_1 \quad & +3e_2 \quad & -2e_3 = 0 \quad & (3) \end{aligned}$$

These 3 equations are the same. So the eigenvector is obtained from the single equation:

$$e_1 - 3e_2 + 2e_3 = 0$$

We can freely choose any two of the components, e_1, e_2 *or* e_3 at will, with the remaining one determined by the above equation.

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Class Exercise 3 - Solution

$$e_1 - 3e_2 + 2e_3 = 0$$

Set $e_2 = \alpha$, $e_3 = \beta$, then $e_1 = 3\alpha - 2\beta$, we get

$$\mathbf{e} = \begin{pmatrix} 3\alpha - 2\beta \\ \alpha \\ \beta \end{pmatrix} = \alpha \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$$

Now ⍰= 2 is an eigenvalue of multiplicity 2, and we seek, if possible, two independent eigenvectors defined by $e_1 - 3e_2 + 2e_3 = 0$.

Setting α=1 and β=0, we get $\mathbf{e}^{(2)} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$.

Setting α=0 and β=1, we get $\mathbf{e}^{(3)} = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$.

Trace of Matrix

The trace of a square matrix is defined as the sum of its diagonal elements, which is always equal to the sum of its eigenvalues.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \longrightarrow Tr(A) = a_{11} + a_{22} + \cdots + a_{nn} = \lambda_1 + \lambda_2 + \cdots + \lambda_n$$

Please verify the above for the previous examples.

Eigendecomposition / Diagonalization

One important application of eigenvalues and eigenvectors is the eigendecomposition (or diagonalization) of a transformation matrix. From the eigenvector equation,

$$\begin{aligned}
A\mathbf{v} &= \lambda\mathbf{v} \\
[A\mathbf{v_1} \quad A\mathbf{v_2} \quad \ldots \quad A\mathbf{v_n}] &= [\lambda_1\mathbf{v_1} \quad \lambda_2\mathbf{v_2} \quad \ldots \quad \lambda_n\mathbf{v_n}] \\
\longrightarrow A\,[\mathbf{v_1} \quad \mathbf{v_2} \quad \ldots \quad \mathbf{v_n}] &= [\mathbf{v_1} \quad \mathbf{v_2} \quad \ldots \quad \mathbf{v_n}] \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{bmatrix} \\
AP &= PD \\
A &= PDP^{-1}
\end{aligned}$$

So matrix A is 'decomposed' into 3 matrices containing eigenvalues or eigenvectors, hence called **eigendecomposition**.

Eigendecomposition / Diagonalization

So, to obtain the eigendecomposition (or diagonalization) of a matrix, we just put the matrix in the form:

$$A = PDP^{-1}$$

$$P = [v_1 \;\; v_2 \;\; \dots \;\; v_n]$$

(matrix of column eigenvectors)

$$D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

(diagonal matrix of eigenvalues)

$$P^{-1} = [v_1 \;\; v_2 \;\; \dots \;\; v_n]^{-1}$$

Eigendecomposition is also known as spectral decomposition. Some matrices are non-diagonalizable and are known as defective matrices (not covered in this course).

Example

Diagonalize the matrix $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

We have found that the eigenvectors are

$$x^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad x^{(2)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix},$$

with eigenvalues 1 and -1 respectively.

Let

$$P = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then

$$P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

We can write $A = PDP^{-1}$.

Example

Diagonalize

$$A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

Example

We have found that the eigenvectors are

$$e^{(1)} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \quad e^{(2)} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}, \quad e^{(3)} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

with eigenvalues 2, 1 and -1 respectively.

Let

$$P = \begin{pmatrix} 1 & 3 & 1 \\ 3 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Then

$$P^{-1} = \frac{1}{6} \begin{pmatrix} -2 & 2 & 2 \\ 3 & 0 & -3 \\ -1 & -2 & 7 \end{pmatrix}.$$

We can write $A = PDP^{-1}$.

Class Exercise

Diagonalize the matrix A.

$$A = \begin{pmatrix} 3 & -3 & 2 \\ -1 & 5 & -2 \\ -1 & 3 & 0 \end{pmatrix}$$

Some observations

- An $n \times n$ matrix **A** gives n number of λ

- The eigenvalues can be:
  - Imaginary, a conjugate pair
  - Real and distinct
  - Complex, also must be a conjugate pair
  - Multiple real and/or multiple pairs of complex conjugates

- Eigenvectors are not unique
  - There are many other choices.
  - However, there are only n independent eigenvectors for an $n \times n$ matrix (if the rank of the matrix is n).

Eigendecomposition / Diagonalization

A matrix in its diagonalized form is easy to manipulate in math operations. Eg, to compute A^2, we have:

$$\begin{aligned} A^2 &= AA = \left(PDP^{-1}\right)\left(PDP^{-1}\right) \\ &= PD\left(P^{-1}P\right)DP^{-1} \\ &= PD(I)DP^{-1} \\ &= PDDP^{-1} = PD^2P^{-1} \end{aligned}$$

Since D is a diagonal matrix, D^2 just means squaring each diagonal element.

By repeating the above, we can get $A^n = PD^nP^{-1}$. So to obtain A raised to power n, simply raise each element of D to the same power.

Eigendecomposition / Diagonalization

Combined with Taylor series, we can get the various functions of A, such as e^A, ln (A), sin(A) etc. For example, we can derive:

From Taylor series for e^x.

$$
\begin{aligned}
e^A &= I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \ldots + \frac{A^n}{n!} + \ldots \\
&= PIP^{-1} + PDP^{-1} + \frac{PD^2P^{-1}}{2!} + \frac{PD^3P^{-1}}{3!} + \ldots + \frac{PD^nP^{-1}}{n!} + \ldots \\
&= P\left(I + D + \frac{D^2}{2!} + \frac{D^3}{3!} + \ldots + \frac{D^n}{n!} + \ldots\right)P^{-1} \\
&= Pe^D P^{-1}
\end{aligned}
$$

The same can be done for other elementary functions of a diagonalizable matrix A. Simply apply the function to each eigenvalue in the diagonal matrix D.

Eigendecomposition / Diagonalization

Example: Diagonalize matrix A. Verify that PDP^{-1} = A. $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$

Solution:

The characteristic polynomial equation: $\det(A - \lambda I) = 0$

Therefore $(3-\lambda)(2-\lambda) = 0.$ λ =2 or 3.

When $\lambda = 2$, $3v_1 + v_2 = 2v_1$ which gives $v_1 + v_2 = 0$. Therefore, $\boldsymbol{v_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

When $\lambda = 3$, $3v_1 + v_2 = 3v_1$ which gives $v_2 = 0$. Therefore, $\boldsymbol{v_2} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Now, $P = [\boldsymbol{v_1} \quad \boldsymbol{v_2}] = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$. $D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$.

And $P^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$

$$PDP^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} = A$$

Eigendecomposition / Diagonalization

Example: From the previous example, compute $7e^A$. $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$

Solution:

We've got, $P = [\boldsymbol{v_1} \quad \boldsymbol{v_2}] = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$. $D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$.

And $P^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$ with of course $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$.

From previous slide, we have

$$e^A = Pe^D P^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} e^{\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}} \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} e^2 & 0 \\ 0 & e^3 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} e^2 & e^3 \\ -e^2 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} e^3 & -e^2 + e^3 \\ 0 & e^2 \end{bmatrix}$$

Therefore, $7e^A = 7 \begin{bmatrix} e^3 & -e^2 + e^3 \\ 0 & e^2 \end{bmatrix}$.

Computation of e^D

$$
\begin{aligned}
e^{D} &= 1 + \frac{D}{1!} + \frac{D^{2}}{2!} + \frac{D^{3}}{3!} + \frac{D^{4}}{4!} + . \ . \ . \quad . \\
&= \sum_{n=0}^{\infty} \frac{D^{n}}{n!} \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} D^{n} = \sum_{n=0}^{\infty} \frac{1}{n!} \begin{bmatrix} \lambda_1^n & 0 & 0 & 0 \\ 0 & \lambda_2^n & 0 & 0 \\ & & & \\ 0 & 0 & 0 & \lambda_n^n \end{bmatrix} \\
&= \begin{bmatrix} \sum_{n=0}^{\infty} \frac{\lambda_1^n}{n!} & 0 & 0 & 0 \\ 0 & \sum_{n=0}^{\infty} \frac{\lambda_2^n}{n!} & 0 & 0 \\ 0 & 0 & 0 & \sum_{n=0}^{\infty} \frac{\lambda_n^n}{n!} \end{bmatrix} = \begin{bmatrix} e^{\lambda_1} & 0 & 0 & 0 \\ 0 & e^{\lambda_2} & 0 & 0 \\ & & & \\ 0 & 0 & 0 & e^{\lambda_n} \end{bmatrix}
\end{aligned}
$$

Optional: Page Ranking Exercise by MATLab

https://matlabacademy.mathworks.com

https://en.wikipedia.org/wiki/PageRank

Page Ranking

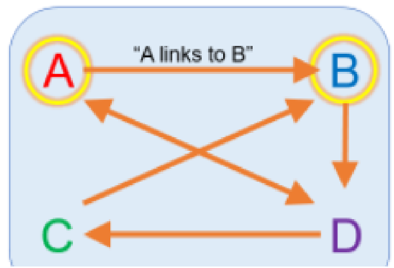
Search engines rank website pages according to the importance of each page. You can use eigenvalue decomposition to find the importance of a page.

In this activity, you will use a simple example of four different websites. The pages are named **A**, B, C, and D. They contain links to each other. For example, page **A** contains links to B and D

Page A

Links to Page B and Page D:
www.pageB.com
www.pageD.com

The relationship between webpages can be viewed as a graph. An arrow between **A** and B means that '**A** contains a link to B'.



If you repeat this operation many times (`A*A*A*...*A*c`), the resulting vector is the dominant eigenvector of `A`.

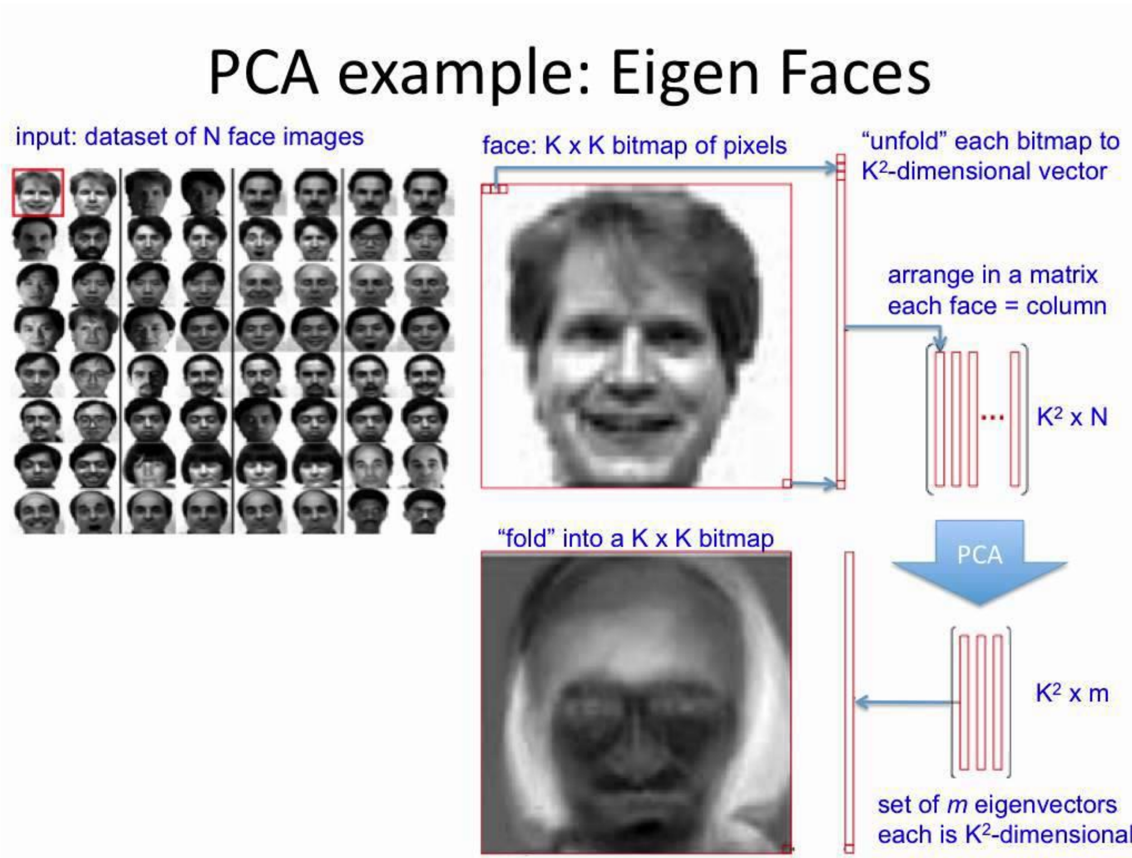
Calculating the dominant eigenvector of `A` will also calculate the probability that a user will be at a certain page after navigating the pages for an infinite amount of time.

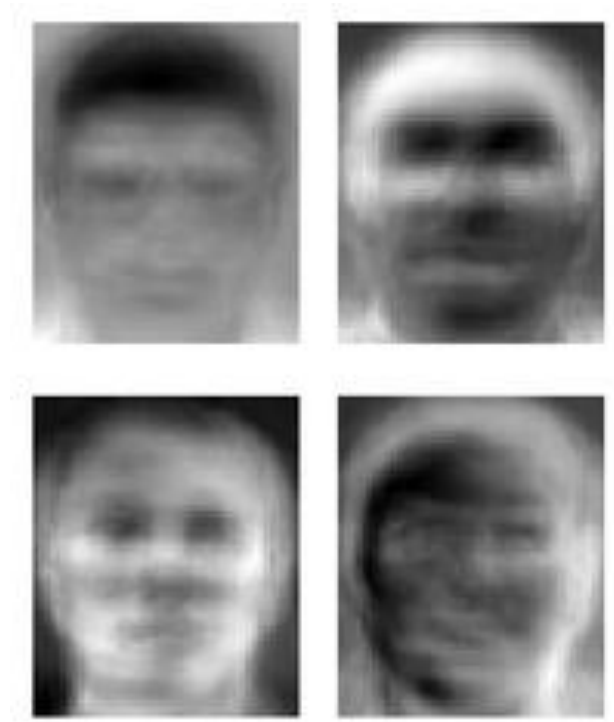
This graph can be represented as a matrix. Each column of the matrix is the probability of a user going from one page to another. Probabilities, and these columns, always sum to one.

$$\begin{array}{cc} & \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \end{array} & \begin{bmatrix} 0.1 & 0 & 0 & 0.45 \\ 0.45 & 0.1 & 0.9 & 0 \\ 0 & 0 & 0.1 & 0.45 \\ 0.45 & 0.9 & 0 & 0.1 \end{bmatrix} \end{array}$$

0 means that a link does not exist between those two pages. Setting the diagonal elements to 0.1 means there is a 10% chance that someone will remain on the same page, instead of clicking on a link.

Eigenfaces (Optional) Optional:



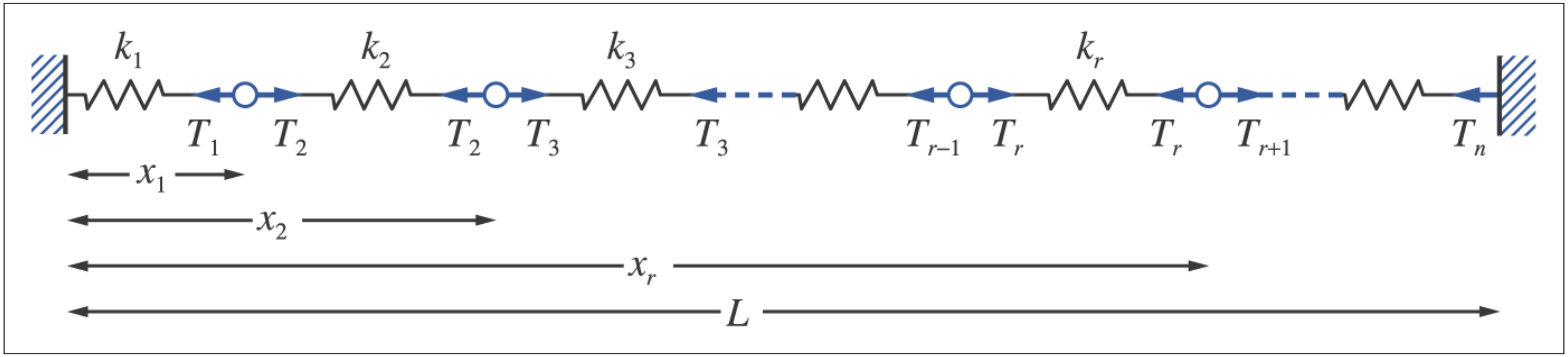


https://www.youtube.com/watch?v=_lY74pXWlS8

https://en.wikipedia.org/wiki/Eigenface

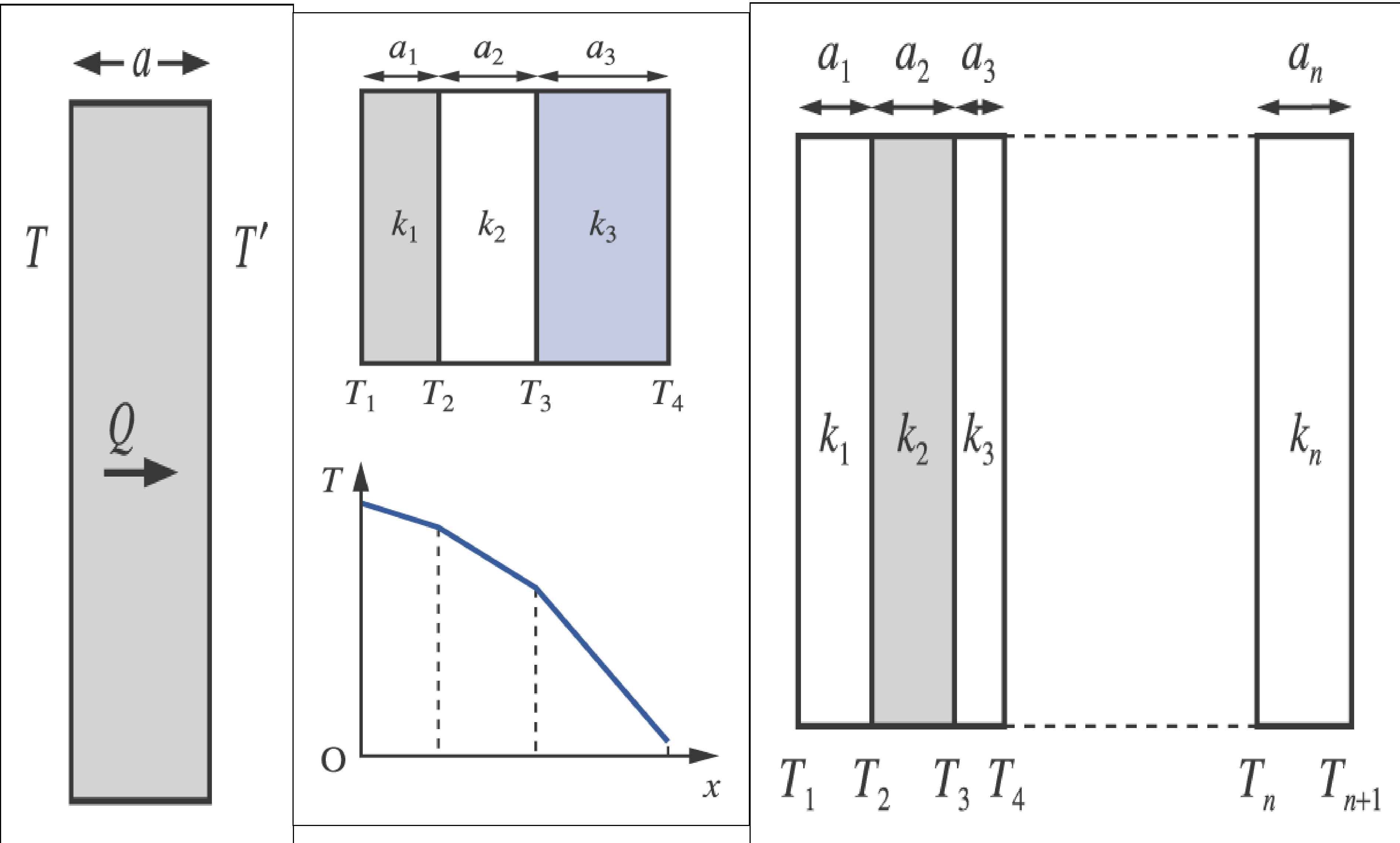
Principal component analysis (**PCA**) is a statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called principal components.

APPLICATIONS from Textbook



n-particle system from textbook

APPLICATIONS from Textbook



Multiple layer of heat transfer from textbook